Important Questions

1. What is a multivalued dependency? Give an example
2. List the four properties of a transaction.
3. What is the purpose of a log in log-based recovery?
4. Differentiate between primary index and secondary index.
5. Differentiate between cascading rollback and recoverable schedules.
6. Explain why a schema in 3NF may not be in BCNF.
7. What is the significance of the order of a B+ Tree in insertion and deletion operations?
8. Define phantom read. How is it different from a non-repeatable read?
9. Why does redundancy lead to anomalies in relational databases?
10. Define the concept of lossless join decomposition with an example.
11. What is recoverability in transactions and why is it necessary?
12. Differentiate between primary index and clustered index.

1. A banking application requires that whenever a withdrawal occurs on a savings account and the balance falls below 500, an alert should be generated.
   (a) Write a SQL trigger for the same.
   (b) Describe how this trigger behaves when a batch update is issued affecting multiple rows. Is the trigger executed once per row or once per statement? Why?

2. You are given the following relation:
Invoice(ItemID, ItemName, SupplierID, SupplierName, Category, CategoryDescription)
Assume the following functional dependencies hold:

   ItemID → ItemName, SupplierID, Category

   SupplierID → SupplierName

   Category → CategoryDescription

a. Identify partial and transitive dependencies in the relation.
b. Normalize the relation to 3NF, showing intermediate steps clearly.

3. You are given the following operations in a B+ Tree of order 3 (maximum 2 keys per node): Insert the keys in this order: 5, 15, 25, 30, 35
   a. Show the B+ Tree after each insertion.
   b. Explain why the leaf nodes are linked in B+ Trees and how this aids performance.

4. A relation R(P, Q, R, S, T) has the following functional dependencies:
   { P → Q, Q → R, RS → T, T → P }
   a. Identify all candidate keys.
   b. Normalize the relation into BCNF with step-wise reasoning.

5. Consider a B+ Tree of order 3 (i.e., each internal node can have at most 3 children). Insert the following keys in order: 5, 18, 22, 36, 10, 20
   (a) Delete key 10 and show the structure of the tree after deletion.
   (b) Mention what happens when underflow occurs in B+ Trees.